AI for Science: Bridging Machine Learning and Scientific Discovery

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October 28, 2025

Feynman's words

What I cannot create, I do not understand.

Outline

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What is AI for Science?

AI for Science represents the intersection of artificial intelligence and scientific research, where machine learning methods are applied to accelerate scientific discovery.

- Drug Discovery: Predicting molecular properties and designing new compounds
- Materials Science: Discovering new materials with desired properties
- Climate Modeling: Understanding and predicting climate patterns
- Physics: Solving complex physical systems and equations
- Biology: Understanding protein folding and biological processes

Key Challenges in AI for Science

Scientific problems present unique challenges for machine learning:

1. Data scarcity: Limited experimental data, expensive to collect
2. Physical constraints: Solutions must obey physical laws
3. Interpretability: Scientific understanding requires explainable models
4. Uncertainty quantification: Scientific predictions need confidence bounds
5. Multi-scale phenomena: From quantum to macroscopic scales

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Physics-Informed Neural Networks

PINNs incorporate physical laws directly into neural network training:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics} \tag{1}$$

where

- $\mathcal{L}_{data}$: Standard data fitting loss
- $\mathcal{L}_{physics}$: Physics constraint loss (PDEs, conservation laws)
- λ: Weighting parameter balancing data and physics

Example: Solving PDEs with PINNs

Consider the heat equation:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u + f(x,t) \tag{2}$$

with boundary conditions $u(x,0) = g(x)$ and $u(\partial\Omega, t) = h(x,t)$.
The physics loss becomes:

$$\mathcal{L}_{physics} = \left\| \frac{\partial u_{NN}}{\partial t} - \alpha \nabla^2 u_{NN} - f \right\|^2 + \lambda_1 \left\| u_{NN}(x,0) - g(x) \right\|^2 + \lambda_2 \left\| u_{NN}(\partial\Omega, t) - h(x,t) \right\|^2 \tag{3}$$

where $u_{NN}(x,t;\theta)$ is the neural network approximation.

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Advantages of PINNs

- No mesh required: Unlike traditional numerical methods
- Unified framework: Handle forward and inverse problems
- Data fusion: Combine sparse data with physical knowledge
- Parameter discovery: Learn unknown parameters in PDEs

Challenges:

- Training can be unstable
- Requires careful hyperparameter tuning
- May struggle with complex geometries

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Molecular Representation Learning

Converting molecules to machine-readable formats:

- SMILES strings: Text-based molecular representation
- Graph neural networks: Molecular graphs with atoms as nodes, bonds as edges
- 3D coordinates: Spatial arrangement of atoms
- Descriptors: Hand-crafted molecular features

Graph Neural Networks for Molecules

For a molecular graph $G = (V, E)$ with atoms $v_i \in V$ and bonds $e_{ij} \in E$:
Message passing:

$$h_v^{(l+1)} = \text{UPDATE}\left(h_v^{(l)}, \text{AGGREGATE}\left(\{h_u^{(l)} : u \in \mathcal{N}(v)\}\right)\right) \tag{4}$$

Property prediction:

$$y = \text{READOUT}(\{h_v^{(L)} : v \in V\}) \tag{5}$$

where L is the number of message passing layers.

Applications in Drug Discovery

- Property prediction: Solubility, toxicity, binding affinity
- Molecular generation: Designing new drug candidates
- Retrosynthesis: Planning synthetic routes
- Protein-ligand binding: Predicting drug-target interactions

Key datasets:

- QM9: 134k small organic molecules; GEOM-Drugs
- ChEMBL: Large-scale bioactivity data
- PDB: Protein structures; AlphaFoldDB

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Climate Modeling with AI

Climate systems involve complex interactions across multiple scales:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F} \tag{6}$$

where $\mathbf{u}$ is velocity, p is pressure, and $\mathbf{F}$ represents external forces.

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Weather Prediction

Traditional approach: Numerical weather prediction (NWP) using physics-based models.

AI approach: Deep learning models trained on historical weather data:

- ConvLSTM: Spatiotemporal weather prediction
- Graph neural networks: Modeling atmospheric dynamics
- Transformer models: Long-range dependencies in weather patterns; ViT; Swin Transformer;

Hybrid approach: Combining AI with traditional NWP for improved accuracy.

Climate Change Detection

Using AI to detect and attribute climate change:

- Pattern recognition: Identifying climate change signatures
- Attribution studies: Linking extreme events to climate change
- Scenario modeling: Predicting future climate under different emissions

Key challenges:

- Limited historical data
- Non-stationary climate system
- Uncertainty quantification

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The Protein Folding Problem

Proteins fold from linear amino acid sequences to 3D structures:

$$\text{Sequence: ACDEFGHIKLMNPQRSTVWY} \rightarrow \text{3D Structure} \tag{7}$$

Key challenges:

- Exponential number of possible conformations
- Complex energy landscapes
- Multiple timescales (nanoseconds to seconds)

AlphaFold and Structure Prediction

AlphaFold approach:

1. Multiple sequence alignment: Evolutionary information
2. Attention mechanisms: Long-range interactions
3. Structure module: 3D coordinate prediction
4. Confidence estimation: Reliability of predictions

Key innovations:

- End-to-end differentiable architecture
- Integration of evolutionary and physical constraints
- Uncertainty quantification

architecture

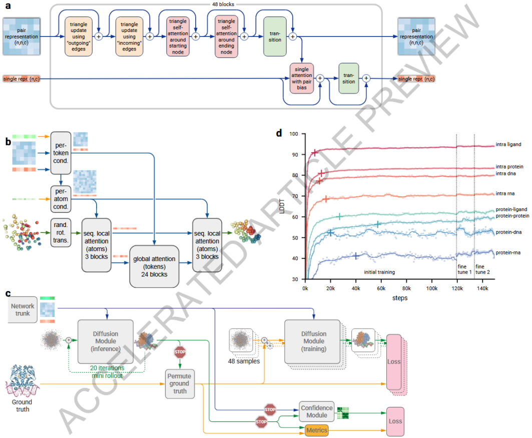

Figure: The architecture of AlphaFold 3.

Impact of AlphaFold

- Structural biology revolution: Millions of protein structures predicted
- Drug discovery: Better understanding of drug targets
- Protein design: Engineering proteins with desired functions
- Evolutionary biology: Understanding protein evolution

Current limitations:

- Limited accuracy for disordered regions
- Challenges with protein complexes
- Difficulty with membrane proteins

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Materials Property Prediction

Predicting material properties from composition and structure:

- Electronic properties: Band gap, conductivity, magnetic properties
- Mechanical properties: Elastic modulus, strength, hardness
- Thermal properties: Heat capacity, thermal conductivity
- Optical properties: Refractive index, absorption spectra

Crystal Structure Representation

Periodic systems: Representing crystal structures with periodic boundary conditions.

Graph representations:

- Crystal graphs: Atoms as nodes, bonds as edges
- Periodic embeddings: Handling translational symmetry
- Multi-scale features: From atomic to macroscopic properties

Deep learning approaches:

- CGCNN: Crystal Graph Convolutional Neural Networks
- SchNet: Quantum-chemical property prediction
- MEGNet: Materials property prediction

High-Throughput Materials Discovery

Computational screening:

1. Generate candidate materials
2. Predict properties using ML models
3. Filter promising candidates
4. Experimental validation

Key databases:

- Materials Project: 150k+ materials with computed properties
- OQMD: Open Quantum Materials Database
- AFLOW: Automatic Flow for Materials Discovery

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Accelerating Scientific Simulations

AI can accelerate traditional scientific computing:

- Surrogate models: Fast approximations of expensive simulations, fast and efficient such as in nuclear fusion
- Multiscale modeling: Bridging different length and time scales
- Parameter optimization: Finding optimal simulation parameters
- Uncertainty quantification: Propagating uncertainties through simulations

Neural Operators

Learning mappings between function spaces:

$$G : \mathcal{U} \to \mathcal{V}, \quad u \mapsto v \tag{8}$$

where $\mathcal{U}$ and $\mathcal{V}$ are function spaces.
Examples:

- Neural ODEs: Learning dynamics of ordinary differential equations
- Neural PDEs: Solving partial differential equations
- Fourier Neural Operators: Learning in frequency domain

Applications in Scientific Computing

- Fluid dynamics: Turbulence modeling, flow prediction
- Quantum chemistry: Electronic structure calculations
- Solid mechanics: Stress analysis, fracture prediction
- Electromagnetics: Wave propagation, antenna design

Benefits:

- Orders of magnitude speedup
- Reduced computational cost
- Real-time predictions

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Current Challenges in AI for Science

1. Data quality and quantity: Limited, noisy, or biased scientific data
2. Physical consistency: Ensuring AI predictions obey physical laws
3. Interpretability: Understanding what AI models learn
4. Generalization: Models that work across different domains
5. Uncertainty quantification: Reliable error estimates

Future Directions

- Foundation models for science: Large-scale models trained on scientific data
- Automated hypothesis generation: AI systems that propose new scientific theories
- Scientific reasoning: AI that can perform logical scientific reasoning
- Multimodal scientific AI: Integrating different types of scientific data
- Scientific collaboration: AI systems that work alongside human scientists

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Ethical Considerations

- Scientific integrity: Ensuring AI doesn't introduce bias in scientific research
- Reproducibility: Making AI for science research reproducible
- Open science: Sharing models, data, and code
- Scientific communication: How AI affects scientific publishing and peer review

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Concept & Setup

Goal. Predict full-atom 3D structures from an amino-acid sequence using a general-purpose Transformer trained with flow matching.

Design stance.

- No MSA, no explicit pair representations, no triangular updates, no equivariant blocks.
- Treat sequence as a conditioning signal (like a text prompt) and generate all-atom coordinates.

Flow-Matching Scheme

Continuous-time generation.

$$\mathrm{d}\mathbf{x}_t = \mathbf{v}_\theta(\mathbf{x}_t, t)\,\mathrm{d}t, \quad t \in [0,1]$$

Map a tractable prior p_0 (Gaussian) to data p_D via a learned velocity field.

Training interpolant (rectified flow).

$$\mathbf{x}_t = t\,\mathbf{x} + (1-t)\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0,\mathbf{I}), \quad \mathbf{x} \sim p_D$$

Target velocity: $\mathbf{v}_t = \mathbf{x} - \epsilon$.
Loss: $\mathbb{E}\big[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}_t\|_2^2\big]$.

Sampling

Stochastic flow sampling. Initialize $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$, integrate to $t = 1$ with Euler–Maruyama:

$$\mathrm{d}\mathbf{x}_t = \mathbf{v}_\theta(\mathbf{x}_t, s, t)\,\mathrm{d}t + \tfrac{1}{2}w(t)\,\mathbf{s}_\theta(\mathbf{x}_t, s, t)\,\mathrm{d}t + \sqrt{\tau\, w(t)}\,\mathrm{d}\bar{\mathbf{W}}_t,$$

where $\mathbf{s}_\theta = \dfrac{t\,\mathbf{v}_\theta - \mathbf{x}_t}{1-t}$ and $w(t) = \dfrac{2(1-t)}{t+\eta}$.
Role of τ. Controls diversity/accuracy trade-off for ensembles.

Folding with Flow Matching

Conditioning. Given sequence s and N_a heavy atoms:

$$\mathbf{x}_t, \mathbf{x}, \boldsymbol{\epsilon} \in \mathbb{R}^{N_a \times 3}, \qquad \mathbf{v}_\theta = \mathbf{v}_\theta(\mathbf{x}_t, s, t)$$

Main objective (per-atom average).

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{\mathbf{x},s,\boldsymbol{\epsilon},t}\left[\frac{1}{N_a}\left\|\mathbf{v}_\theta(\mathbf{x}_t, s, t) - (\mathbf{x} - \boldsymbol{\epsilon})\right\|_2^2\right]$$

Structural term (LDDT-inspired). One-step Euler estimate $\hat{\mathbf{x}}(\mathbf{x}_t) = \mathbf{x}_t + (1-t)\mathbf{v}_\theta(\mathbf{x}_t, s, t)$ and penalize local distance errors:

$$\mathcal{L}_{\mathrm{LDDT}} = \mathbb{E}\left[\frac{\sum_{i \neq j} \mathbf{1}(\delta_{ij} < C)\, \sigma\left(\|\delta_{ij} - \hat{\delta}_{ij}^{t}\|\right)}{\sum_{i \neq j} \mathbf{1}(\delta_{ij} < C)}\right]$$

Total loss: $\mathcal{L} = \mathcal{L}_{\mathrm{FM}} + \alpha(t)\mathcal{L}_{\mathrm{LDDT}}$.
Timestep resampling. $p(t) = 0.98\,\mathrm{LogisticNormal}(0.8, 1.7) + 0.02\,\mathcal{U}(0, 1)$, biased toward $t \to 1$ for fine details.

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Architecture Overview

Three modules, one block type.

1. Atom encoder (local attention): noisy coords $\mathbf{x}_t$ + atom features $\rightarrow$ atom tokens $\mathbf{a} \in \mathbb{R}^{N_a \times d_a}$.
2. Residue trunk (heavy): group/pool atom tokens to residue tokens $\mathbf{r} \in \mathbb{R}^{N_r \times d_a}$, concat with frozen PLM embeddings $e \in \mathbb{R}^{N_r \times d_e}$ from ESM2-3B, process with standard Transformer blocks with adaptive layers.
3. Atom decoder (local attention): ungroup residue tokens back to atoms, add skip from encoder, predict velocity $\hat{\mathbf{v}}_t$.

Blocks. QK-normalization, SwiGLU FFN, adaptive scale/shift by t; all non-equivariant, no pair/triangle stacks.

Positional encodings. RoPE in trunk; 4D axial RoPE in atom modules (3D reference conformer coords + residue index).

Design Choices vs Prior Work

- No MSA, no explicit pair representation, no triangular updates.
- Learn SO(3) symmetries via data augmentation rather than equivariance constraints.
- Hierarchical compute: fine (atoms) $\rightarrow$ coarse (residues) $\rightarrow$ fine (atoms).
- Frozen PLM (ESM2-3B) supplies sequence conditioning; folding model is trained end-to-end with flow matching.

Confidence Module (pLDDT)

What. A separate 4-layer Transformer predicts per-residue pLDDT (0–100).
How.

- Freeze the folding model; feed generated $\hat{\mathbf{x}}$ with $t=1$ to get final residue tokens.
- Train pLDDT head with cross-entropy over 50 bins.

Use. Correlates with LDDT-Cα and can score structures from other models.

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Training Data & Regime

Data sources.

- PDB (cutoff May 2020) $\sim$160k structures.
- AFDB SwissProt subset (pLDDT mean $>$85, std $<$15) $\sim$270k.
- AFESM representatives filtered at pLDDT > 0.8 $\sim$1.9M.
- For 3B model: extended AFESM-E up to $\sim$8.6M distilled structures (max 10 per cluster, mean pLDDT > 80).

Training.

- Pre-train at seq length 256; $\alpha(t) = 1$; AdamW $1\text{e}{-4}$; EMA 0.999; large effective batch.
- Finetune on PDB+SwissProt at seq length 512; $\alpha(t) = 1 + 8\,\mathrm{ReLU}(t - 0.5)$ up to 5 near $t{=}1$.

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Structure and Results

Type	Model	TM-score ↑	GDT-TS ↑	LDDT ↑	LDDT-C_α ↑	RMSD ↓
	CAMEO22					
MSA-based	RoseTTAFold (Baek et al., 2021)	0.780 / 0.860	0.715 / 0.775	0.575 / 0.605	0.798 / 0.827	5.721 / 2.864
	AlphaFlow (Jing et al., 2024a)	0.840 / 0.927	0.808 / 0.853	0.741 / 0.798	0.855 / 0.893	3.846 / 2.122
	AlphaFold2 (Jumper et al., 2021)	0.863 / 0.942	0.844 / 0.903	0.816 / 0.856	0.893 / 0.923	3.578 / 1.857
	RoseTTAFold2 (Baek et al., 2023)	0.864 / 0.947	0.845 / 0.904	0.727 / 0.767	0.893 / 0.926	3.571 / 1.707
PLM-based	ESM3 (Hayes et al., 2025)	0.746 / 0.840	0.694 / 0.758	–	–	–
	ESMDiff (Lu et al., 2024a)	0.754 / 0.847	0.701 / 0.760	–	–	–
	EigenFold (Jing et al., 2023)	0.750 / 0.840	0.710 / 0.790	–	–	–
	OmegaFold (Wu et al., 2022)	0.805 / 0.899	0.767 / 0.844	0.746 / 0.815	0.829 / 0.892	5.294 / 2.622
	ESMFlow (Jing et al., 2024a)	0.818 / 0.893	0.774 / 0.832	0.696 / 0.745	0.827 / 0.867	4.528 / 2.693
	ESMFold (Lin et al., 2023b)	0.853 / 0.933	0.826 / 0.875	0.792 / 0.834	0.871 / 0.906	3.973 / 2.019
Ours	SimpleFold-100M	0.803 / 0.878	0.746 / 0.787	0.721 / 0.752	0.822 / 0.852	4.897 / 2.855
	SimpleFold-360M	0.826 / 0.905	0.782 / 0.841	0.773 / 0.803	0.844 / 0.878	4.775 / 2.681
	SimpleFold-700M	0.829 / 0.915	0.788 / 0.845	0.775 / 0.809	0.850 / 0.886	4.557 / 2.423
	SimpleFold-1.1B	0.833 / 0.924	0.793 / 0.851	0.776 / 0.807	0.850 / 0.883	4.350 / 2.334
	SimpleFold-1.6B	0.835 / 0.916	0.799 / 0.864	0.782 / 0.816	0.853 / 0.889	4.397 / 2.187
	SimpleFold-3B	0.837 / 0.916	0.802 / 0.867	0.773 / 0.802	0.852 / 0.884	4.225 / 2.175
	CASP14					
MSA-based	RoseTTAFold (Baek et al., 2021)	0.654 / 0.678	0.562 / 0.572	0.464 / 0.456	0.705 / 0.723	9.676 / 6.420
	AlphaFlow (Jing et al., 2024a)	0.740 / 0.812	0.661 / 0.711	0.632 / 0.662	0.767 / 0.799	7.091 / 3.949
	RoseTTAFold2 (Baek et al., 2023)	0.802 / 0.881	0.740 / 0.824	0.638 / 0.669	0.824 / 0.869	6.744 / 3.292
	AlphaFold2 (Jumper et al., 2021)	0.845 / 0.907	0.783 / 0.855	0.778 / 0.817	0.856 / 0.897	5.027 / 3.015
PLM-based	ESMDiff (Lu et al., 2024a)	0.521 / 0.499	0.447 / 0.430	–	–	–
	ESM3 (Hayes et al., 2025)	0.534 / 0.567	0.459 / 0.488	–	–	–
	EigenFold (Jing et al., 2023)	0.590 / 0.637	0.539 / 0.575	–	–	–
	ESMFlow (Jing et al., 2024a)	0.627 / 0.679	0.539 / 0.544	0.525 / 0.539	0.669 / 0.730	10.503 / 6.974
	OmegaFold (Wu et al., 2022)	0.693 / 0.773	0.625 / 0.723	0.627 / 0.726	0.715 / 0.824	9.845 / 4.042
	ESMFold (Lin et al., 2023b)	0.701 / 0.792	0.622 / 0.711	0.637 / 0.705	0.725 / 0.802	8.679 / 4.016
Ours	SimpleFold-100M	0.611 / 0.628	0.513 / 0.544	0.537 / 0.549	0.659 / 0.685	11.157 / 8.976
	SimpleFold-360M	0.674 / 0.758	0.585 / 0.654	0.617 / 0.657	0.703 / 0.762	9.382 / 4.828
	SimpleFold-700M	0.680 / 0.767	0.591 / 0.668	0.630 / 0.674	0.714 / 0.763	9.289 / 4.431
	SimpleFold-1.1B	0.697 / 0.796	0.607 / 0.668	0.640 / 0.676	0.723 / 0.758	9.249 / 4.462
	SimpleFold-1.6B	0.712 / 0.801	0.630 / 0.709	0.660 / 0.699	0.741 / 0.798	8.424 / 4.722
	SimpleFold-3B	0.720 / 0.792	0.639 / 0.703	0.666 / 0.709	0.747 / 0.829	7.732 / 3.923

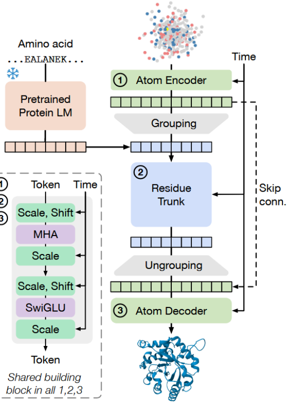

Figure 2 Overview of SimpleFold's architecture built on general-purpose standard Transformer block with adaptive layers. Atom encoder, residue trunk, and atom decoder all share the same general-purposed building block. Our model circumvents the need for pair representations or triangular updates.

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Results

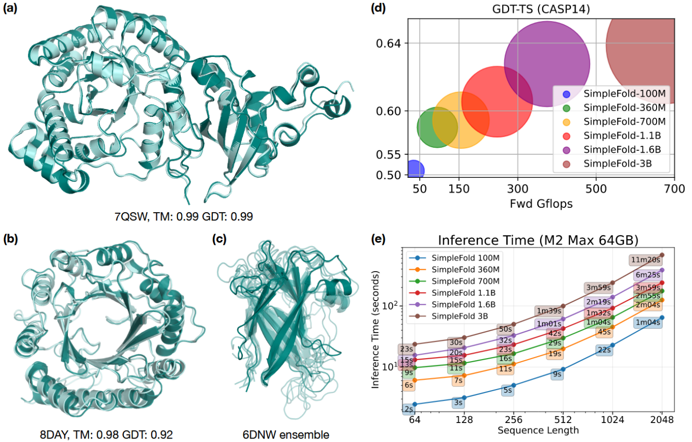

Figure 1 Example predictions of SimpleFold on targets (a) chain A of 7QSW (RubisCO large subunit) and (b) chain A of 8DAY (Dimethylallyltryptophan synthase 1), with ground truth shown in light aqua and prediction in deep teal. (c) Generated ensembles of target chain B of 6NDW (Flagellar hook protein FlgE) with SimpleFold finetuned on MD ensemble data. (d) Performance of SimpleFold on CASP14 with increasing model sizes from 100M to 3B. (e) Inference time of different sizes of SimpleFold on consumer level hardware, i.e., M2 Max 64GB Macbook Pro.

Scalability

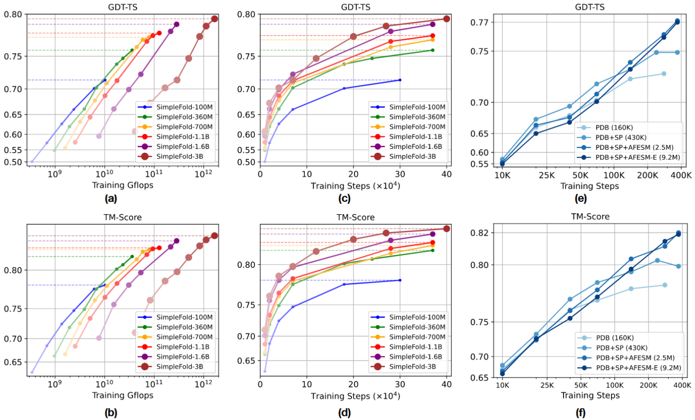

Figure 4 Scaling behavior of SimpleFold. Training Gflops vs. folding performance on GDT-TS and (b) TM-score. Training steps vs. folding performance on (c) GDT-TS and (d) TM-score. How data scale affects the performance (e) GDT-TS and (f) TM-score. All models are benchmarked on CAMEO22.

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AlphaFold 3

See the pdfs.

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UniGenX

See the MS PowerPoint presentation for more details.

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Summary

AI for Science represents a paradigm shift in how we approach scientific research:

- Acceleration: Faster discovery and development cycles
- Integration: Combining data-driven and physics-based approaches
- Automation: Reducing manual work in scientific research
- Insights: Discovering patterns humans might miss

The Future of AI for Science

- Democratization: Making advanced scientific tools accessible to all researchers
- Interdisciplinary collaboration: Breaking down silos between scientific fields
- Scientific method evolution: How AI changes the scientific process
- Grand challenges: Tackling humanity's biggest scientific problems

Questions for Discussion

1. How can we ensure AI for Science maintains scientific rigor and reproducibility?
2. What are the implications of AI for the future of scientific careers?
3. How should we balance AI automation with human scientific intuition?
4. What ethical guidelines should govern AI for Science research?

Welcome inputs

Any comments?

Welcome your inputs!